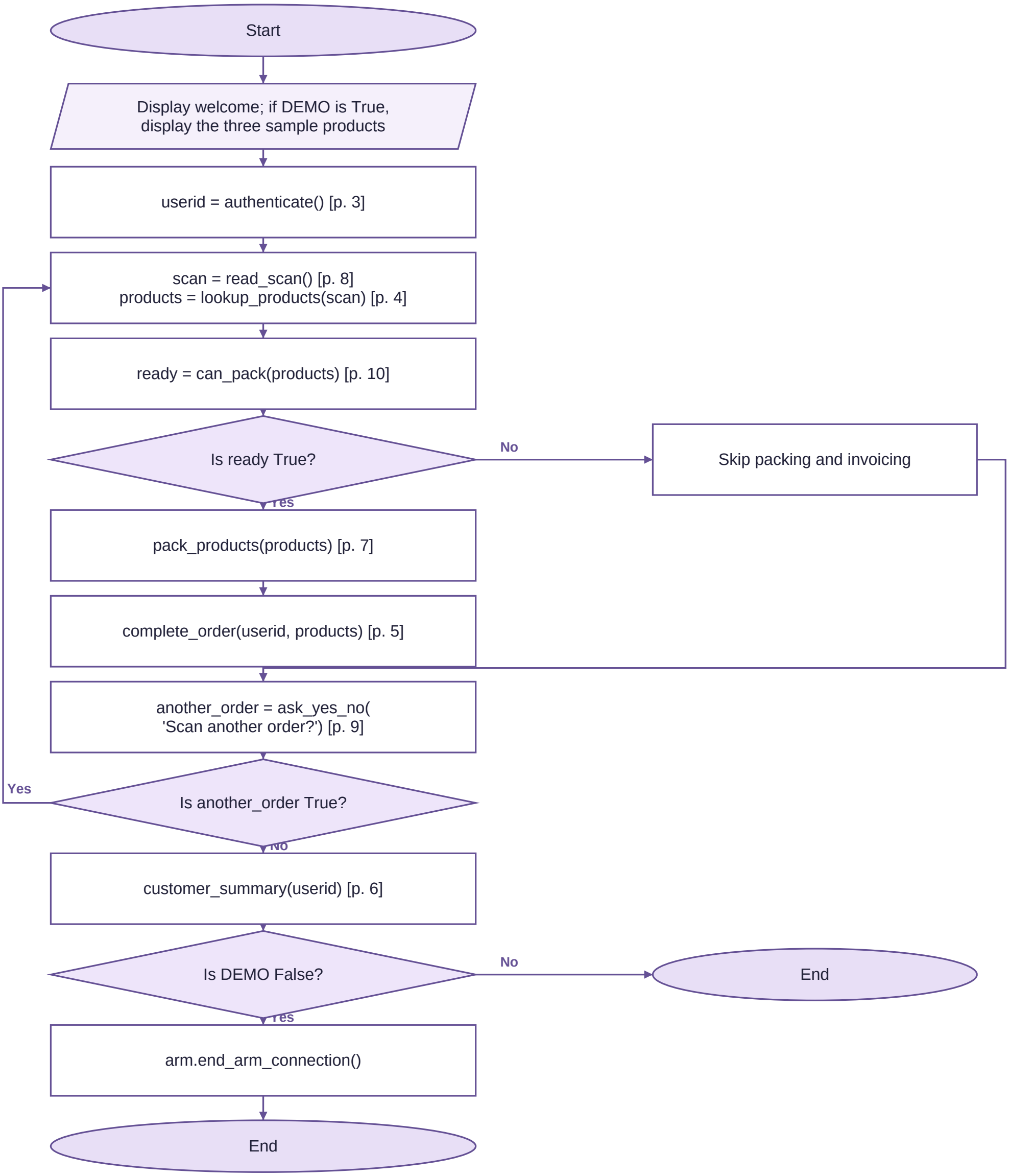
main()

Responsibility Team

Parameters None

Returns None



FUNCTION DETAILS

Each scan is priced once. The same products list goes to packing and invoicing.
can_pack prints the reason for any skipped order. A valid order is packed before it is recorded.

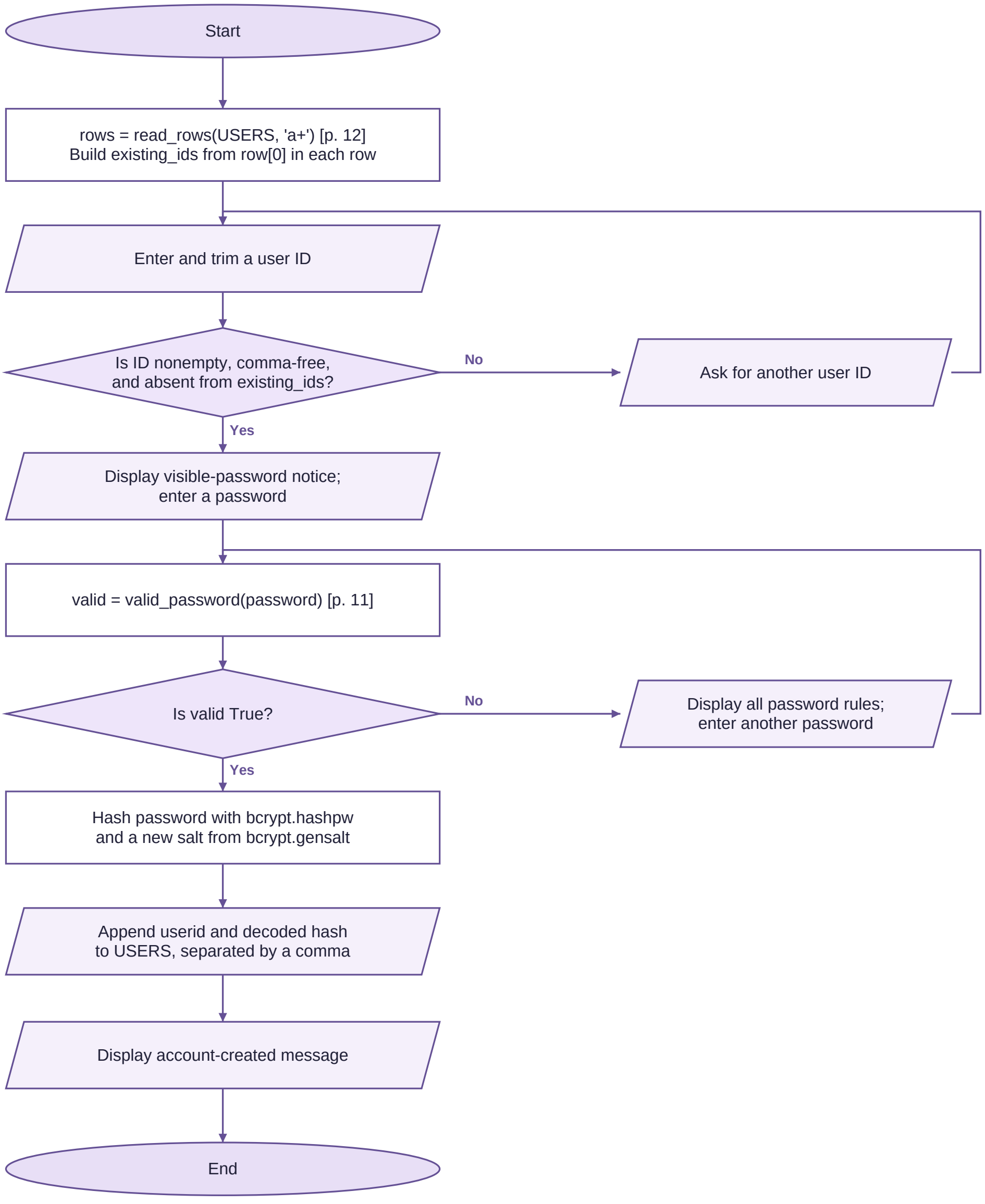
Oval Start / End | Parallelogram Input / Output | Rectangle Process | Diamond Decision

sign_up()

Lead Anshi Wang | Support Joleen Sadek | MacID wanga275

Parameters None

Returns None



FUNCTION DETAILS

USERS is users.csv in the lab and demo_users.csv during laptop practice.
The password helper checks length and all four character requirements. Only the hash is stored.

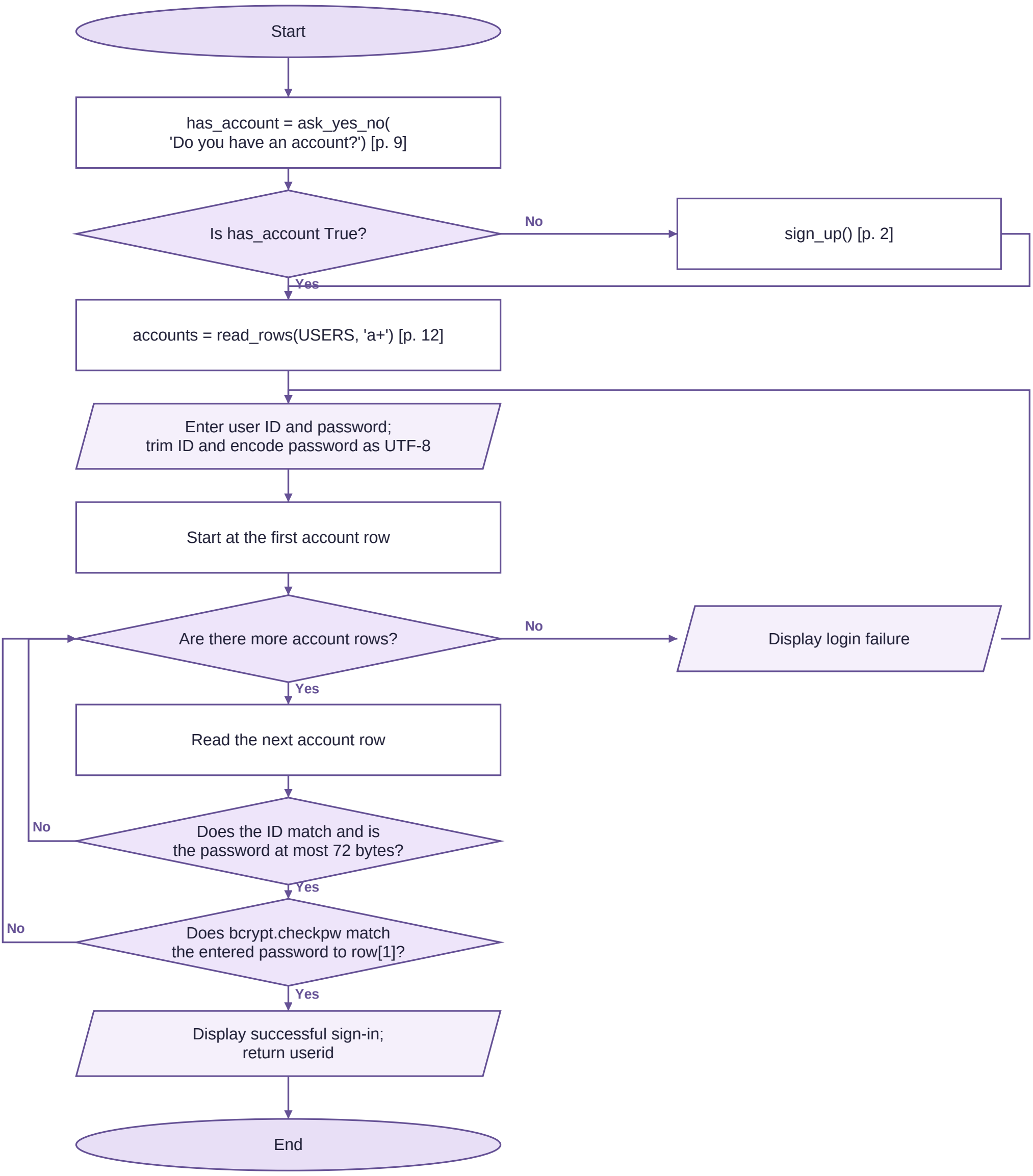
Oval Start / End | Parallelogram Input / Output | Rectangle Process | Diamond Decision

authenticate()

Lead Anshi Wang | Support Margarette Pangan | MacID wanga275

Parameters None

Returns userid (string)



FUNCTION DETAILS

A failed attempt repeats the credential prompt. A successful attempt returns the matching ID.
Every account row is checked before a failure message is shown. Passwords remain hashed in USERS.

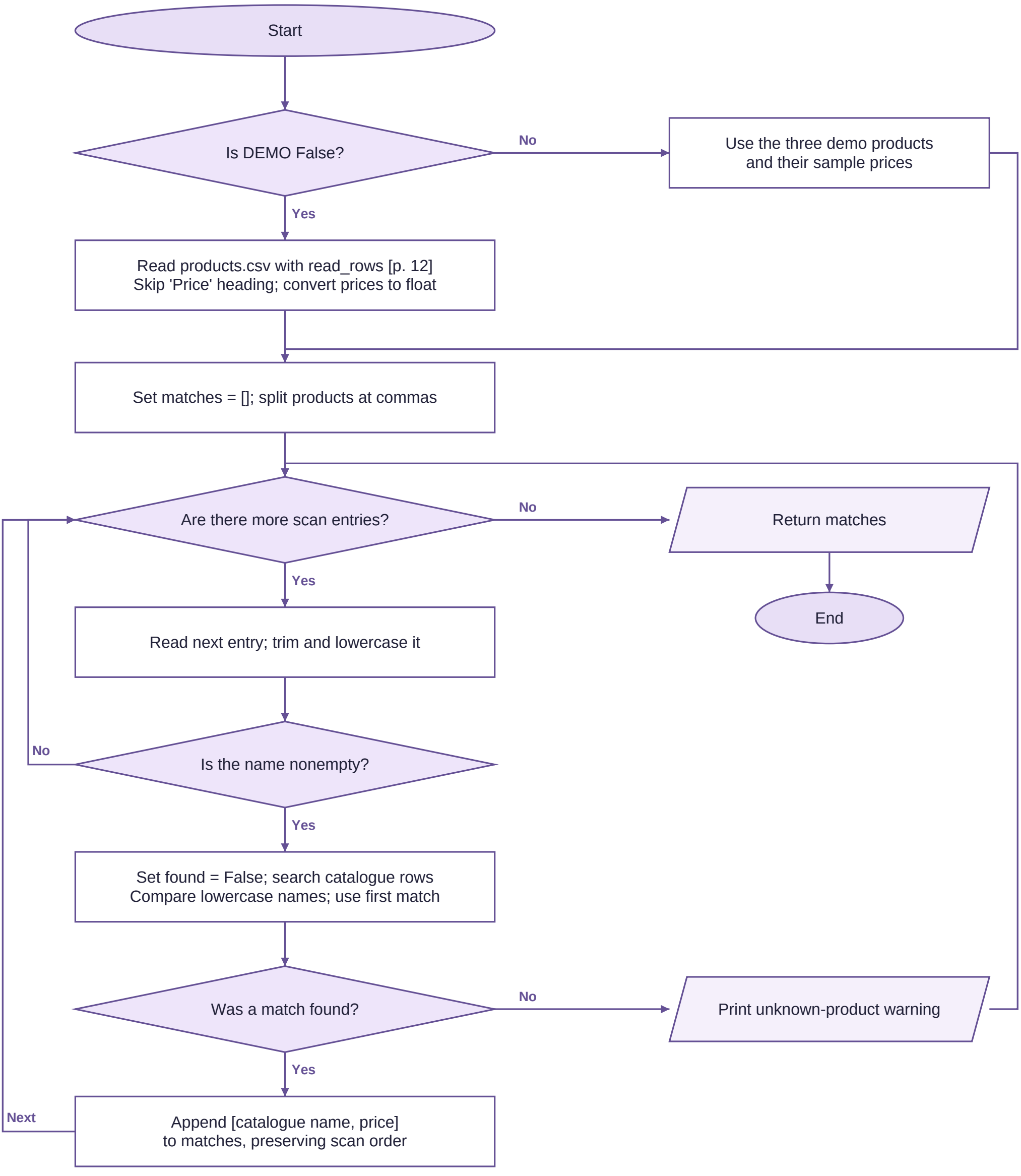
Oval Start / End | Parallelogram Input / Output | Rectangle Process | Diamond Decision

lookup_products(products)

Lead Margarette Pangan | Support Anshi Wang | MacID pangam1

Parameters products (comma-separated string)

Returns [[name, numeric price], ...]



FUNCTION DETAILS

Search each scanned name independently. Repeated names produce repeated product pairs.
Unknown names are warned about and skipped. Empty entries are skipped. No matches returns [].

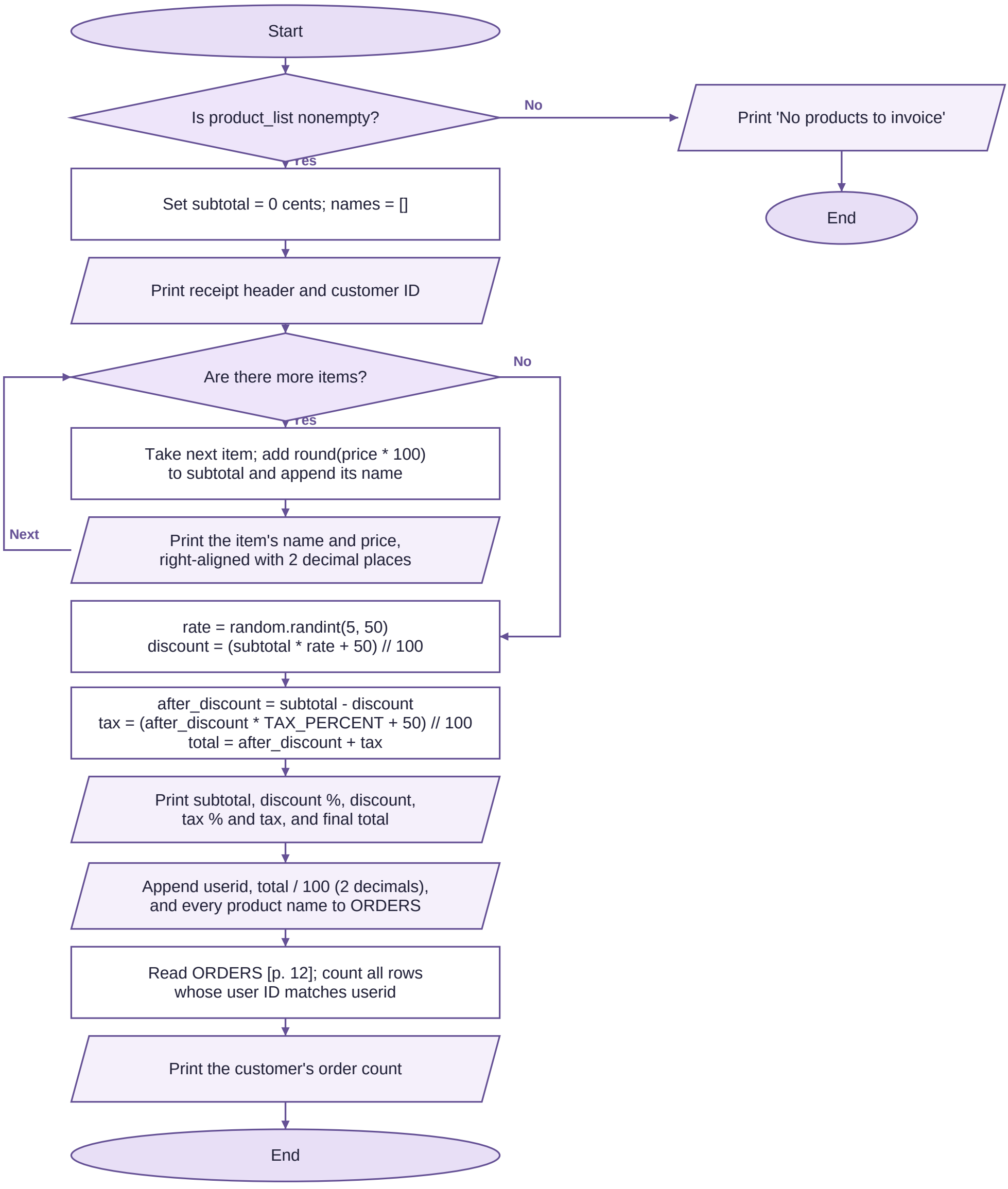
Oval Start / End | Parallelogram Input / Output | Rectangle Process | Diamond Decision

complete_order(userid, product_list)

Lead Jose Mella Iturra | Support Anshi Wang | MacID mellaiti

Parameters userid; [[name, price], ...]

Returns None



FUNCTION DETAILS

Process each item in order, then calculate the discount and tax for the entire order.
Amounts are calculated in cents; +50 before //100 rounds halves up. TAX_PERCENT = 13 is an assumption.

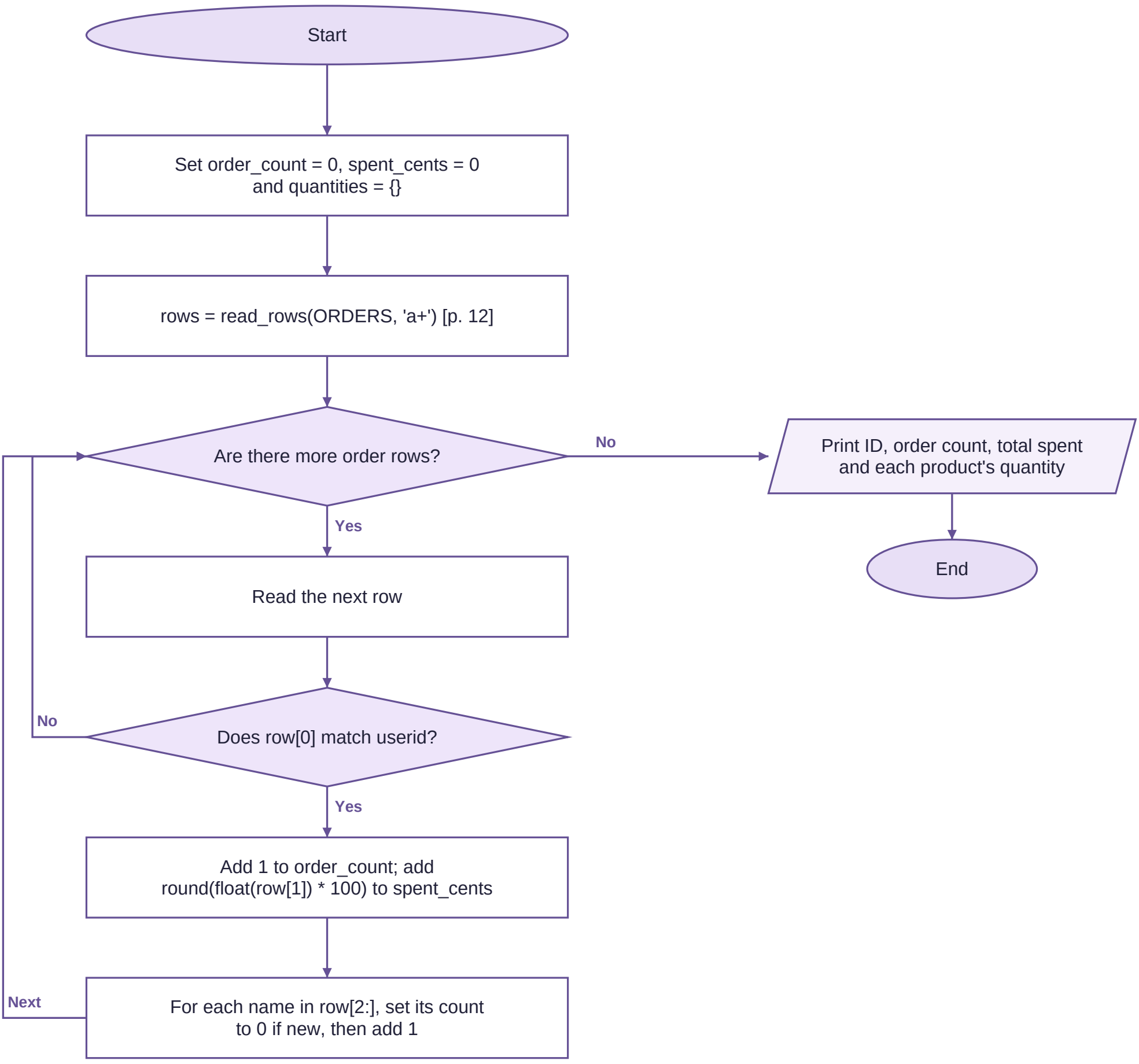
Oval Start / End | Parallelogram Input / Output | Rectangle Process | Diamond Decision

customer_summary(userid)

Lead Joleen Sadek | Support Jose Mella Iturra | MacID sadekj3

Parameters userid

Returns None



FUNCTION DETAILS

Totals include only this user. Each occurrence counts once, including duplicates within an order.
Print money to 2 decimals and quantities right-aligned. With zero orders, also print 'No orders yet'.

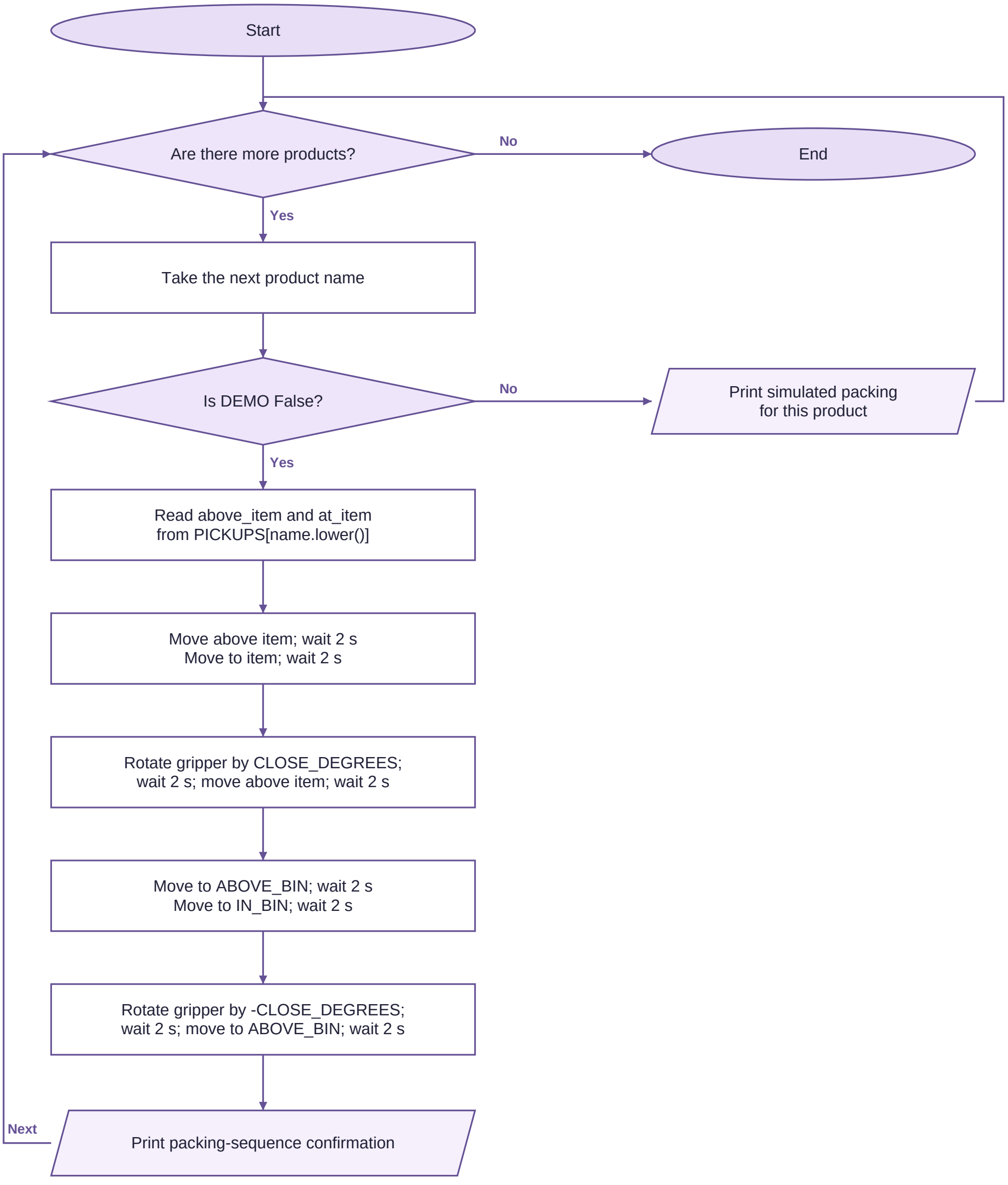
Oval Start / End | Parallelogram Input / Output | Rectangle Process | Diamond Decision

pack_products(product_list)

Responsibility Team

Parameters [[name, price], ...]

Returns None



FUNCTION DETAILS

main calls can_pack before this function. Lab positions and the closing rotation must be supplied. Start with the gripper open. Coordinates are metres. Gripper rotations are relative motor angles.

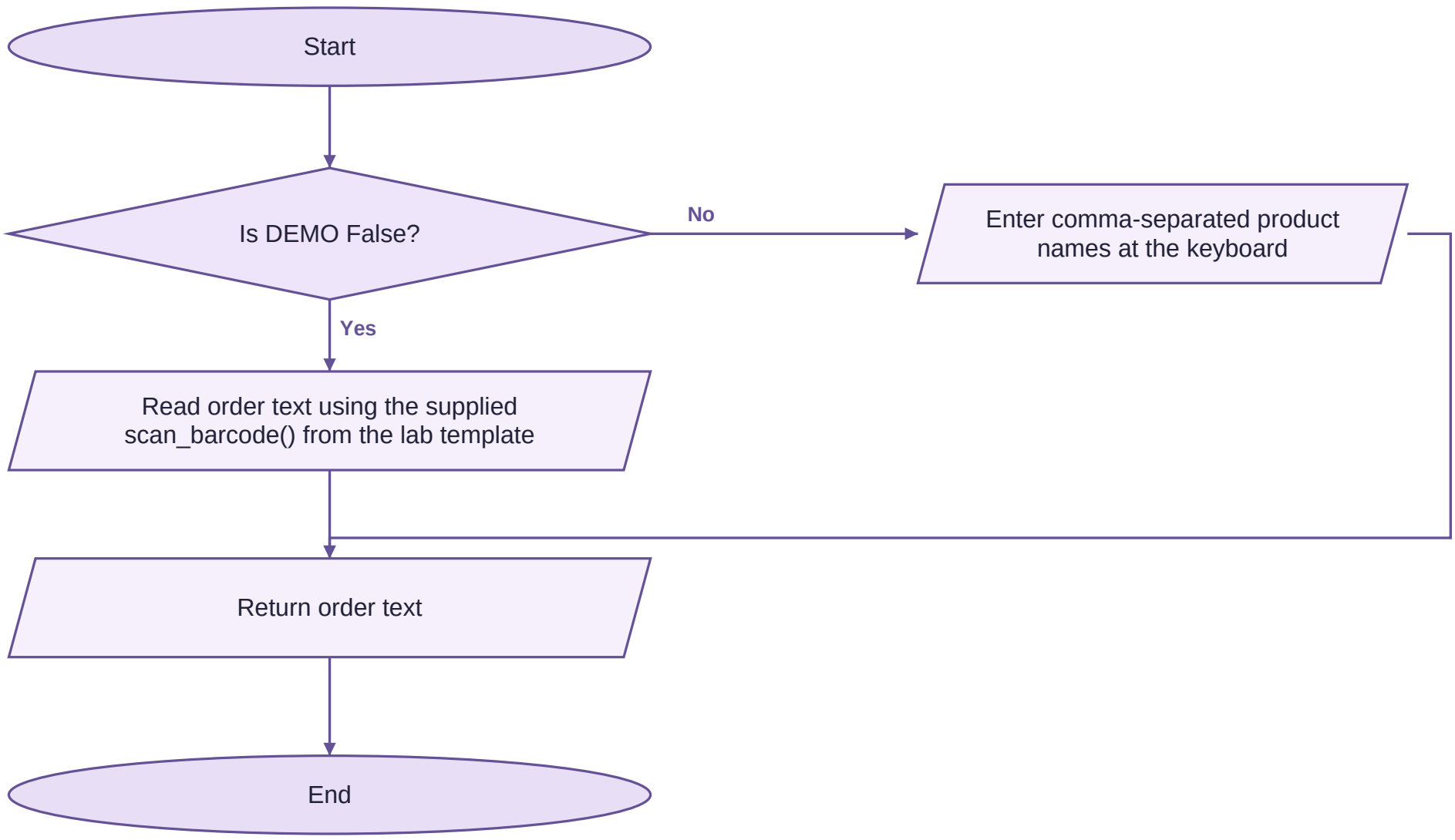
Oval Start / End | Parallelogram Input / Output | Rectangle Process | Diamond Decision

read_scan()

Responsibility Team helper; supports the required functions

Parameters None

Returns scan text (string)



FUNCTION DETAILS

This helper selects how the order is entered. The course supplies scan_barcode().
Laptop example: widget,gear,widget. The resulting string is passed to lookup_products().

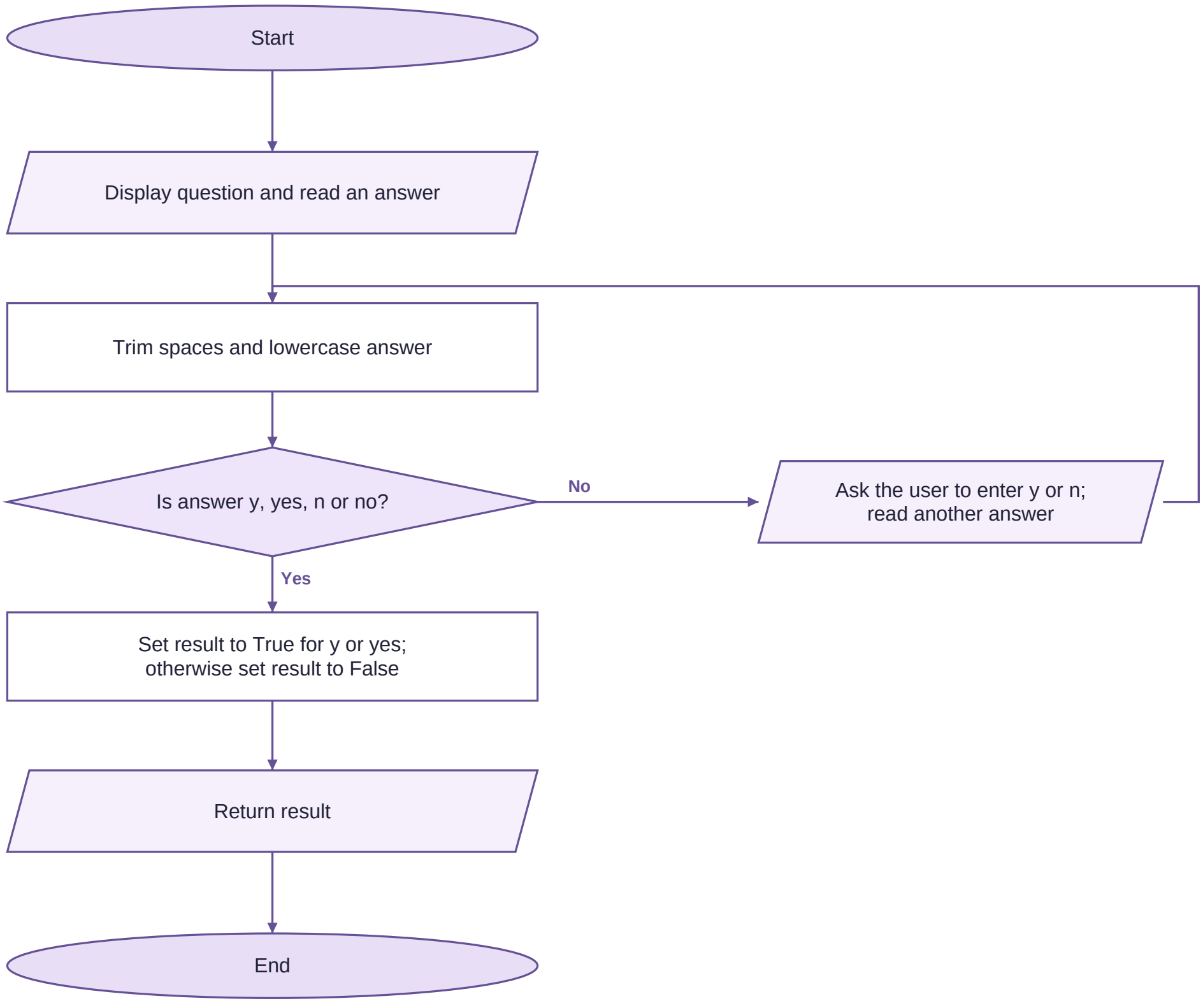
Oval Start / End | Parallelogram Input / Output | Rectangle Process | Diamond Decision

ask_yes_no(question)

Responsibility Team helper; supports the required functions

Parameters question (string)

Returns True or False



FUNCTION DETAILS

Used when asking whether the customer has an account and whether to scan another order.
The answer is validated before the caller receives a Boolean result.

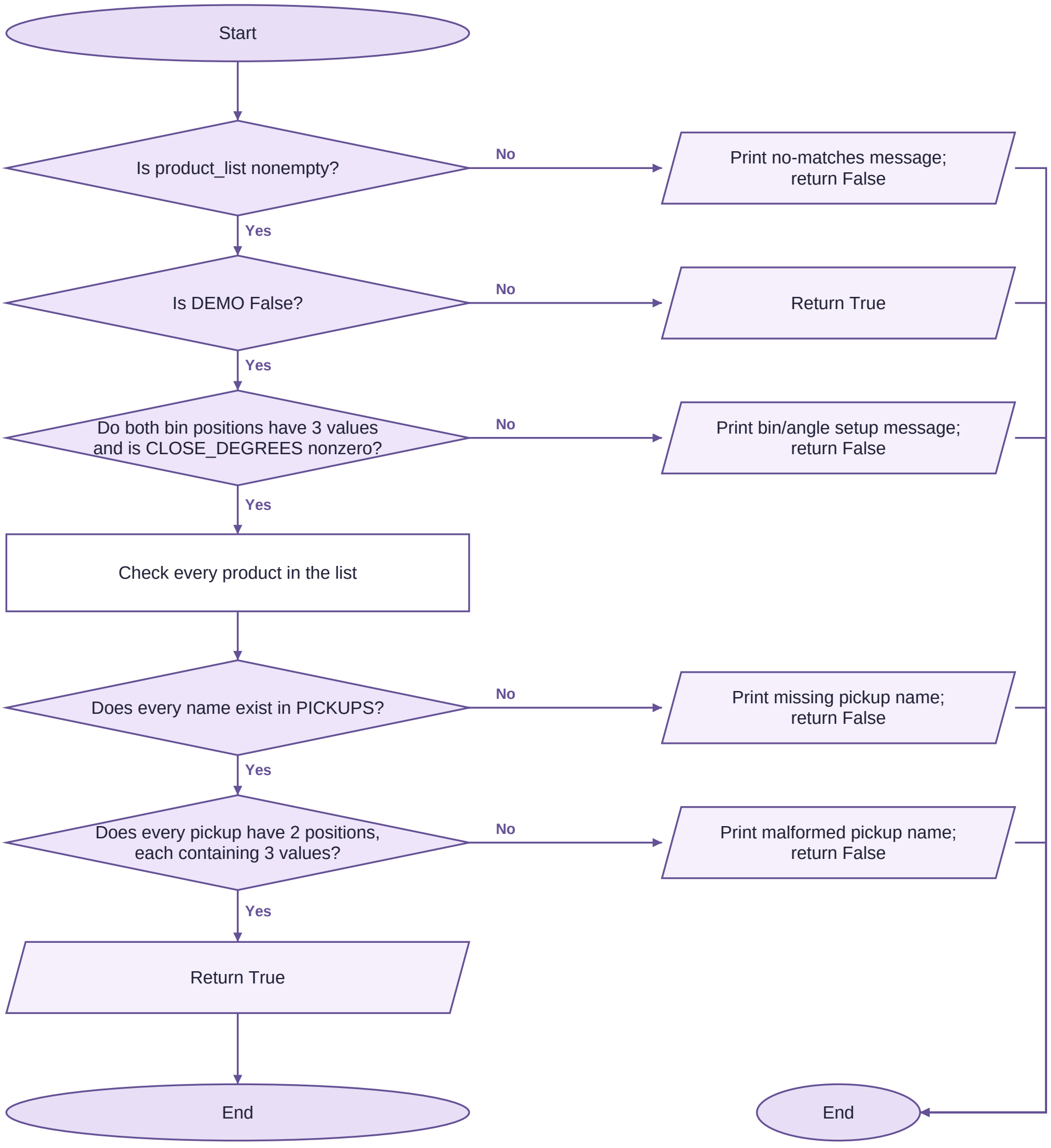
Oval Start / End | Parallelogram Input / Output | Rectangle Process | Diamond Decision

can_pack(product_list)

Responsibility Team helper; supports the required functions

Parameters [[name, price], ...]

Returns True or False



FUNCTION DETAILS

Per product, check its name first, then its two positions. Stop at the first problem.
A False result prevents packing and invoicing. This helper checks structure, not physical calibration.

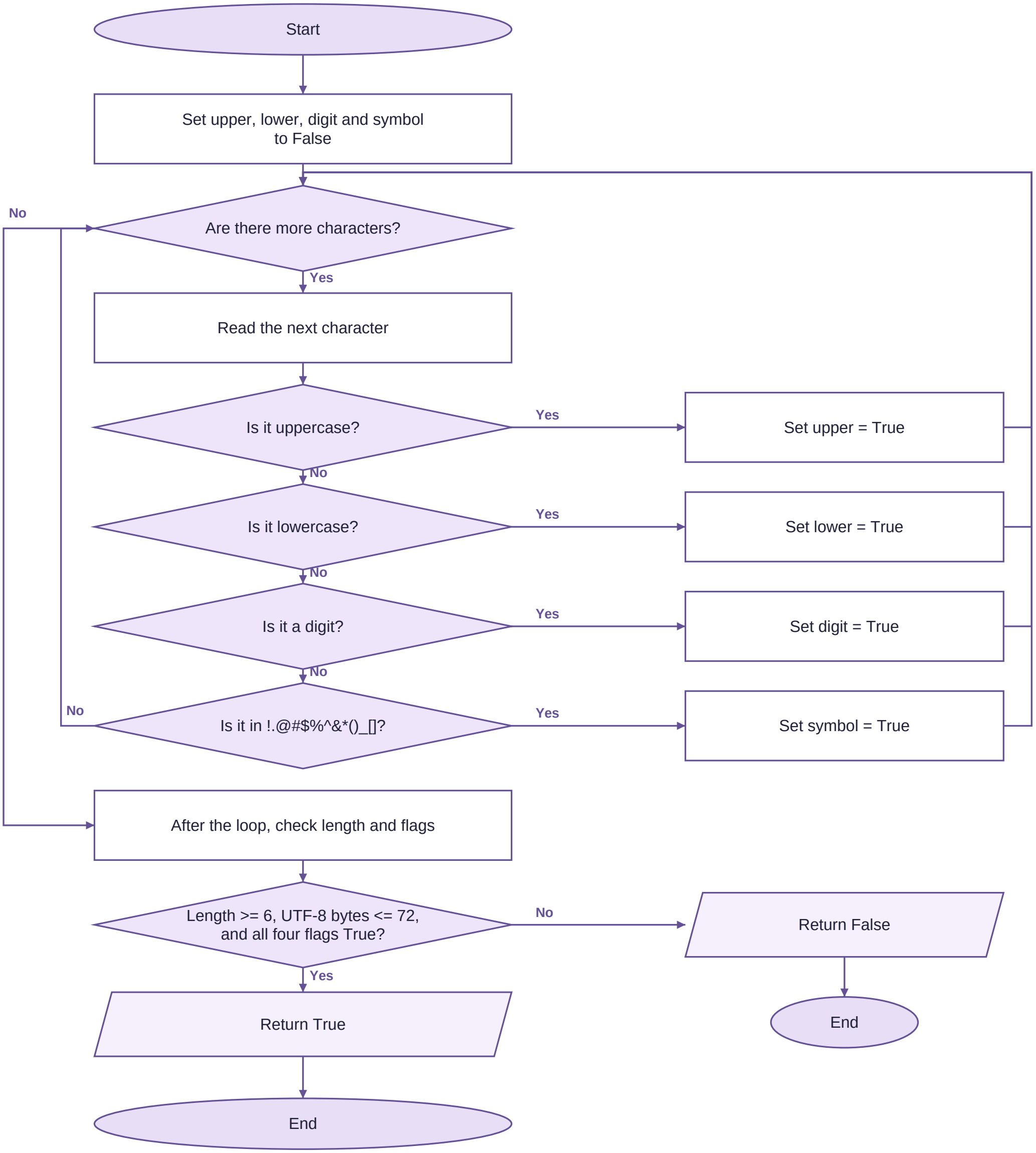
Oval Start / End | Parallelogram Input / Output | Rectangle Process | Diamond Decision

valid_password(password)

Responsibility Team helper; supports the required functions

Parameters password (string)

Returns True or False



FUNCTION DETAILS

This helper supports Anshi Wang's sign_up function. All four flags must be True.
The first matching character test sets its flag. A nonmatching character changes no flags.

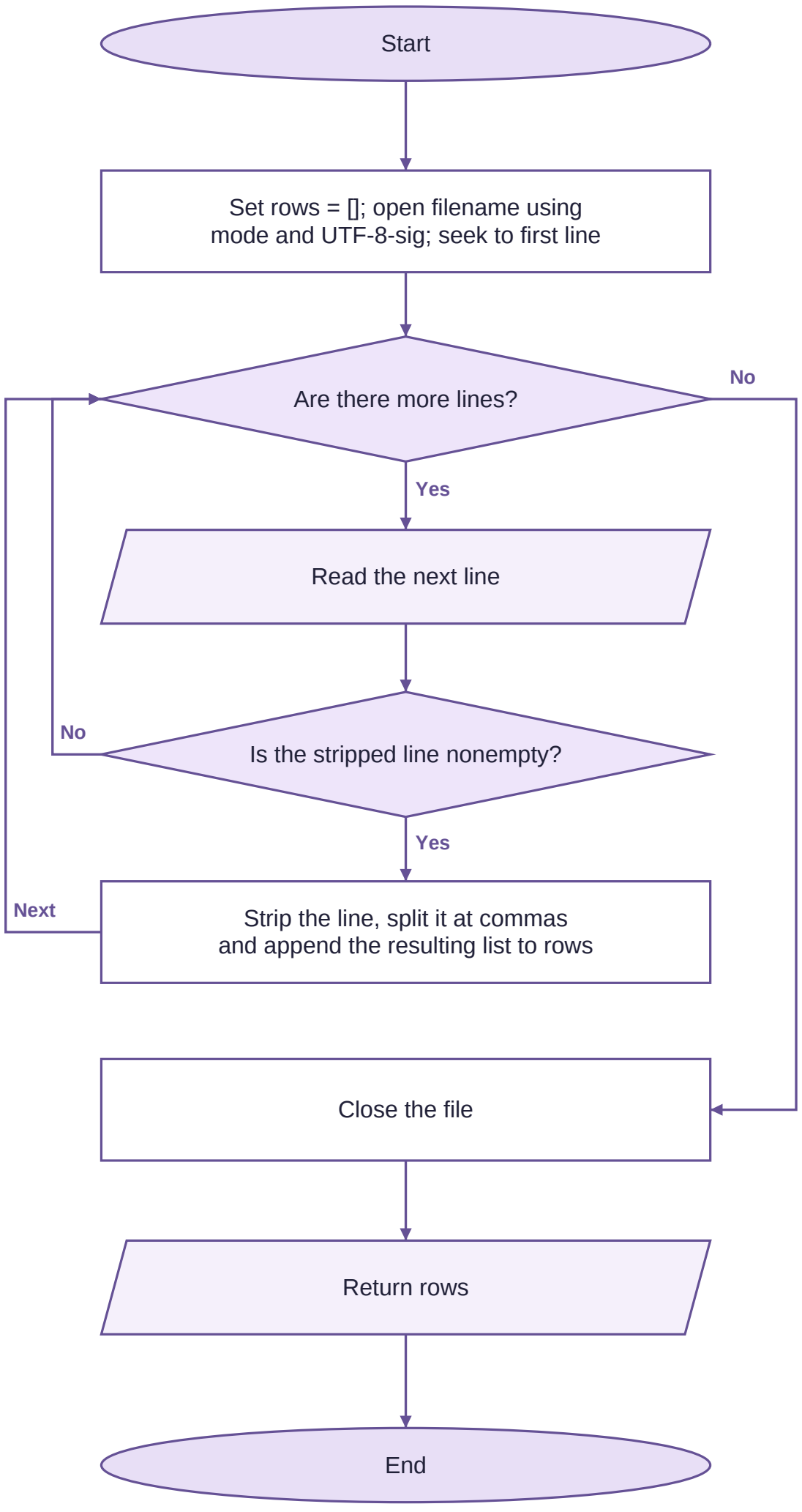
Oval Start / End | Parallelogram Input / Output | Rectangle Process | Diamond Decision

read_rows(filename, mode)

Responsibility Team helper; supports the required functions

Parameters filename; mode (default 'r')

Returns list of rows, each a list of strings



FUNCTION DETAILS

For users and orders, mode 'a+' creates a missing file. products.csv is opened for reading. Blank lines are ignored. The course data format uses comma-separated fields.

Oval Start / End | Parallelogram Input / Output | Rectangle Process | Diamond Decision